

OPIOID ANALGESICS

What are the opioid analgesics?

The drugs that produce drowsiness, mood changes, mental clouding while, they produce their analgesic effects.



The receptors and corresponding action: Opioids act on specific kind of receptors that present in high concentration in CNS (limbic system, thalamus, hypothalamus, brain stem, corpus striatum, and spinal cord) – and other tissues. They are G-protein coupled receptors.

1. **μ (mu):** $\mu 1$ (supraspinal analgesia) & $\mu 2$ (spinal analgesia, euphoria, respiratory depression, sedation, miosis and $\downarrow$ GIT motility).
2. **K(kappa):** spinal analgesia, dysphoria, hallucination, with less miosis and respiratory depression than μ ones.
3. **δ (delta):** the effects in human are: analgesia and release of growth hormone.
4. **Sigma:** are no longer considered to be opioid receptors because their activation is not reversed by the opioid inverse-agonist naloxone).

N.B: Another receptor of clinical importance is the opioid-receptor-like receptor 1 (**ORL1**), which is involved in pain responses as well as having a major role in the development of tolerance to μ -opioid agonists.

Classification of opioid analgesics (according to efficacy):

	Class	Examples
1	Strong agonists	Morphine, fentanyl, pethidine, methadone, heroin
2	Moderate agonist	Codeine, oxycodone
3	Weak agonist	Propoxyphene
4	Mixed agonist-antagonist and partial agonist	Nalbuphene, buprenorphine
5	Antagonist	Naloxone, naltrexone
6	Others	Tramadol, tapentadol

Classification of opioid analgesics (according to nature)

	Class	Examples
1	Natural	Morphine, Codeine
2	Semisynthetic	Heroin, oxymorphone
3	Synthetic	Fentanyl, pethidine, methadone, pentazocine

1.Strong Agonists

A) Morphine



Pharmacological Actions:

1- CNS:

- ✓ *Analgesia*: visceral & somatic dull continuous pain without interference with motor activity or sensations.
- ✓ *Euphoria and Hypnosis*: sensation of wellbeing & mental clouding.
- ✓ *Pupil*: miosis & ↓IOP.
- ✓ *Respiration*: depress all phases of respiratory activity, why?? **Return to lecture**
- ✓ *Cough*: Why?? **See the lecture note for more details**
- ✓ *Nausea & vomiting*: affects the CTZ more in ambulated patients.

2- CVS: Therapeutic doses may produce orthostatic hypotension and fainting due to peripheral VD and histamine release.

3- GIT: It ↓ the motility but ↑ the tone of the smooth Ms. All the sphincters are contracted and the secretions are reduced. Constipation is induced. - The pressure is increased in the biliary tract due to spasm of sphincter of Oddi....ttt by atropine, naloxone and vasodilators.

4- Bronchi: Therapeutic doses produce bronchospasm only in asthmatics, large doses generally cause this.

5- Urinary: Bladder and Uterus: The same effect as in the smooth Ms of the GIT leads to urine retention.

6- Metabolism: It has a transient hyperglycemic effect; it also increases the release of ADH and it inhibits the release of ACTH.

Pharmacokinetics

- ◆ PO (limited oral bioavailability), Transdermal, Rectal, I.V.
- ◆ Metabolized in the liver and eliminated in urine.
- ◆ 2-4 hr half life

Therapeutic Uses

1. **Analgesia:** In sever pains as in terminal malignancy, acute pericarditis, myocardial infarction. In acute cases, it is used by IV or SC routes. In shock, SC route is not recommended.

2. **Pre-anesthetic Medication:** or during general anesthesia via inhalation or iv route, and in large dose as only anesthetic agent especially in cardiac surgeryto↓ the myocardial depression.

3. **Antitussive and Antidiarrheal:** In dry cough – but in case of diarrhea the meperidine congeners are preferred to decrease the addiction possibility.

4. Pulmonary Edema: In LVF it decreases fear and decrease cardiac shock.

Precautions & Contraindications

1. In asthmatic patients.
2. During labor. It prolongs labor and causes neonatal asphaxia.
3. Patients with diminished respiratory reservoir. As obesity, kyphoscoliosis, emphysema or chronic obstructive airway diseases.
4. In cases of head injuries.
5. Age extremities.
6. Acute abdomen and diagnosed biliary colic.
7. Myxedema (low BMR).
8. Reduced blood volume as with hemorrhage.
9. Sever hepatic and/ or renal failure.



Side Effects

1. GIT: nausea, vomiting, constipation and $\uparrow$ biliary pressure.

2. Respiratory: respiratory depression and hypoxia.
3. CVS: orthostatic hypotension.
4. CNS: mental clouding, dizziness, dysphoria, but the most important ones are:
 - **Idiosyncrasy**: A usual dose is ranged from skin rashes to anaphylactoid reactions that enhanced by its histamine release.
 - **Tolerance** to analgesic, euphorogenic, sedative, and respiratory depression effects, but not for GIT or miotic effects (pharmacodynamic tolerance).

Physical Dependence (Addiction):

- Addiction is usually resulted from the euphoric effect of morphine.
- The common features of morphine addict are: skin flushing, euphoria, malnourished and economic problems.
- The signs & symptoms of withdrawal syndrome:
 1. Autonomic hyperactivity (rhinorrhea, hyperpyrexia, vomiting, abdominal cramps, diarrhea, lacrimation, sweating, dilated pupils, increase heart rate).
 2. Behavioral hyperexcitability (anxiety, restlessness, insomnia, and tremors).
 3. Joints and muscles pains.

Treatment of morphine addiction

(Cross dependence with Methadone. It has less withdrawal symptoms), clonidine (it suppress autonomic hyperactivity) and lofexidine may be also used. It is α_2 agonist but with less hypnosis than clonidine.

Acute Morphine Poisoning:

- ✓ Causes: Toxic and lethal doses in normal, pain free adult. The doses increase by addiction.
- ✓ Features include: Respiratory distress, pin point pupil, cyanosis, hypothermia, hypotension and coma.
- ✓ What is the acute morphine poisoning triad?
 - **CNS depression (coma)**
 - **Respiratory depression (cyanosis)**
 - **Pupillary miosis**
- ✓ **Treatment:** i.v. **NALOXONE** to overcome the respiratory depression.

B) Fentanyl



- ✓ It is one of meperidine congeners
- ✓ It has 100 time analgesic effect and respiratory depression as morphine.
- ✓ Its main use is in anesthesia (rapid onset and short duration of action).
- ✓ Fentanyl is combined with local anesthetics to provide epidural analgesia for labor and postoperative pain.

- ✓ It is used in the form of transmucosal (**fentanyl lollipop**) or transdermal patch in cancer patient pain.
- ✓ Its congeners as **alfentanil** and **sufentanil** are very potent analgesics that are used in general anesthesia and post-operative analgesia.

Side effects of fentanyl:

- Chest pain or discomfort.
- Blurred vision
- Difficult or troubled breathing: irregular, fast or slow, or shallow breathing.
- Lightheadedness, dizziness, or fainting.
- Pale or blue lips, fingernails, or skin.
- Severe muscle stiffness and rigidity due to its effect on *dopaminergic transmission*.
- Slow or irregular heartbeat.
- Unusual tiredness.



C) Meperidine (Pethidine)



- ✓ It is about 1/10 as potent as morphine as analgesic (not 1st line opioid analgesic).
- ✓ No cough depression effect so it has no antitussive activity.
- ✓ It has spasmogenic (GIT, UT, gall bladder) less than that of morphine and rarely produces constipation.
- ✓ No miosis, even causes mydriasis, tachycardia due to its antimuscarinic action.
- ✓ It well absorbed from all routes.
- ✓ Its metabolite, Normeperidine accumulates specially with renal insufficiency, causing neurotoxicity which may leads to seizures. So, it is **not suitable for long term** management of pain.
- ✓ It accumulates in elderly and renal failure patients causing toxicity that is not completely antagonized by Naloxone which even may increases seizures.
- ✓ Meperidine should not be used with MAOIs or serotonin reuptake inhibitors (causes **Serotonin Syndrome**)

Therapeutic uses:

It is used in all situations where any other opioid analgesic used but it has no anti-diarrheal or anticough actions.

Side effects and contraindications:

- The same as morphine but urine retention and constipation are less common.
- Convulsions may occur in addicts.
- Physical dependence and withdrawal symptoms are like those of morphine with less autonomic effects, they develop more rapid and shorter duration. ttt is the same (1mg methadone / 20mg meperidine).

Diphenoxylate

It is a derivative of meperidine and is available in combination with atropine (**Lomotil**) that is used in ttt of diarrhea in a recommended dose (10 – 20mg / day) to avoid the morphine like effects as euphoria and physical dependence.

Loperamide (Imodium)

It is similar to diphenoxylate and is used in control diarrhea mainly traveler diarrhea.



D) Methadone

- It is more effective orally (90% P.P.B).
- It has longer half-life than morphine causing cumulative toxicity.
- Tolerance and physical dependence develop slowly but withdrawal symptoms are milder than morphine.
- It is mainly used in analgesia and ttt of opioid abstinence syndrome

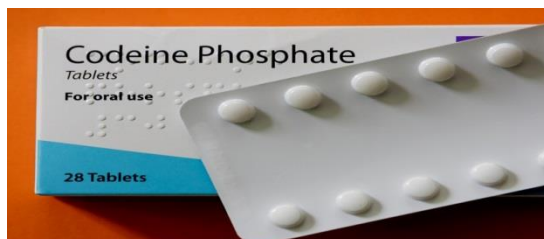
E) Heroin

- ✓ It has rapid onset of action and more potent action than morphine.
- ✓ It has high addiction liability and so not used clinically.

2. Moderate agonists

A) Codeine (Methyl morphine)

- 10% is converted to morphine which responsible for its analgesic activity.
- It is used in mild pain (1/12 analgesic potency of morphine).
- It is mainly a powerful cough depressor with action on specific codeine receptors (it has low addiction and withdrawal liability).



B) Oxycodone

- **Oxycodone** is a semi-synthetic opioid used medically for treatment of moderate to severe pain. It is highly addictive and is a commonly abused drug.
- Common side effects include euphoria, constipation, nausea, vomiting, loss of appetite, drowsiness, dizziness, itching, dry mouth, and sweating. Side effects may also include addiction and dependence.
- It has roughly 1.5 times the effect of the equivalent amount of morphine.

3.MIXED AGONIST- ANTAGONIST AND PARTIAL AGONISTS

- ✓ They mainly act as analgesic and are either κ agonist or μ antagonist as Pentazocine, Nalbuphine, Butorphanol (Stadol), or partial agonist as Buprenorphine.

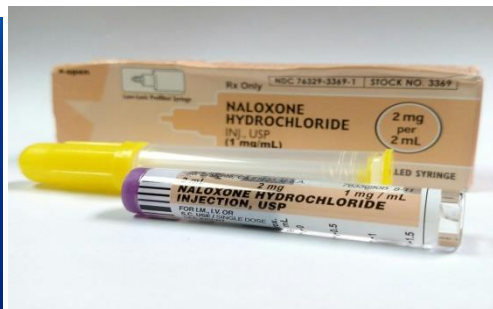
- ✓ Unlike other opioids, it has little to no capacity for euphoria or respiratory depression
- ✓ Side effects include sedation, sweating, vomiting, and dry mouth.
- ✓ They produce analgesia more or less similar to morphine.
- ✓ Pentazocine can produce withdrawal symptoms in morphine addicts, but Butorphanol (Stadol) does this with methadone users. (*Antagonist*)



- ✓ Buprenorphine is about 25-50 times more potent analgesic than morphine and has slower onset, longer duration of action, less withdrawal symptoms.

5.OPIOID ANTAGONISTS

Naloxone: It has a pure antagonistic activity with short duration of action (1-4 h), and given by injection. FDA approved naloxone auto-injector pen and naloxone nasal spray



USES:

1. Treatment of acute opioid poisoning (0.2mg I.V.).

2. Treatment of neonatal opioid-induced fetal asphyxia.
3. Diagnosis of opioid addiction (In addicts lead to withdrawal symptoms and in non addicts lead to nothing).

NALTREXONE: It is a pure antagonist and is used orally, more potent than naloxone and has longer duration of action reached to 24 hours. It may cause hepatotoxicity.



USES:

1. Can be used in ex-abusers to opioids after complete drug withdrawal to block the self- administration of opioids.
2. Has benefit in persons who have frequent access to opioid drugs.
3. Treatment of alcoholism and ethanol addiction.

5.Others

A) Tramadol (mixed opioid/non opioid)

- Codeine analog (μ agonist), inhibits action of NE & 5HT reuptake in CNS.
- Partially antagonized by naloxone.
- It is used on moderate to moderately severe pain, including pain after surgery and diabetic neuropathy. It may cause seizures as a side effect. It considered a Step 2 analgesic

under the World Health Organization's guidelines for the treatment of patients with cancer pain.

N.B: Tramadol is an atypical opioid why?? [Return to lecture](#)



- Low-dose tramadol is a generally safe, effective, and fast-acting antidepressant for a substantial number of individuals suffering from depression.
- The most serious adverse effect of tramadol is cardiac arrhythmia (causes Serotonin Syndrome).

B) Tapentadol

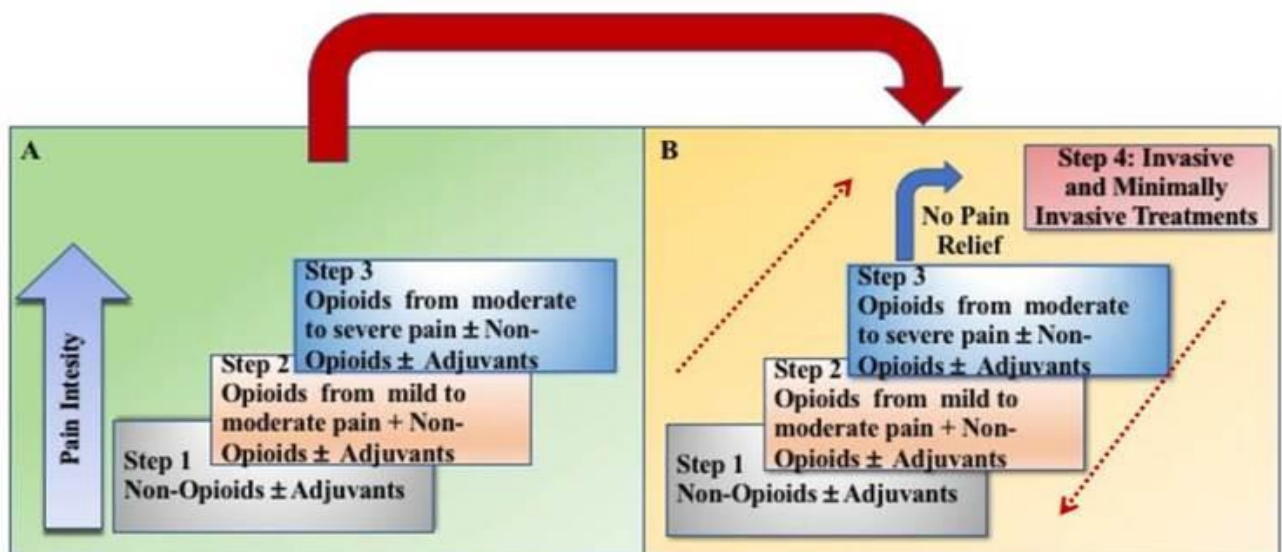
- Compared to tramadol and other opioids, tapentadol is associated with fewer adverse effects than tramadol. Tapentadol is a new alternative to treat acute, chronic, and neuropathic pain



Drug Interactions of opioid

- ✓ Opioid +CNS depressants (sedatives, antidepressants, antipsychotics): add CNS depress.
- ✓ Pethidine or tramadol +MAOIs: serotonin syndrome
- ✓ Partial agonist + pure agonist: withdrawal syndrome and decrease analgesic effect

Analgesic ladder



Transition from the original WHO three-step analgesic ladder (A) to the revised WHO fourth-step form (B). The additional step 4 is an “interventional” step and includes invasive and minimally invasive techniques. This updated WHO ladder provides a bidirectional approach.